

COMP2511

Tute04



Agenda

- SOLID
- Law of Demeter
- Liskov Substitution Principle
- Patterns!!!! YAY!!!!
- Strategy Pattern
- Streams & Lambdas



SOLID

The Principles Of Software Design

- **S**RP (Single Responsibility Principle)
- **O**CP (Open Closed Principle)
- **L**SP (Liskov Substitution Principle)
- **I**SP (Integration Segregation Principle)
- **D**IP (Dependency Inversion Principle)



Law of Demeter

A method should only communicate with its "immediate friends" and not "friends of friends". Method M of object O can use methods of:

- O.
- parameters of M.
- any object created in M.
- O's direct fields.



Law Of Demeter In Action?

X

DON'T: Take apart your calculator, get the CPU and provide electrical signals to the CPU

O

DO: Use the calculator's buttons to make the calculations instead

Liskov Substitution Principle

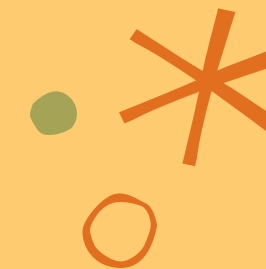
Objects of a superclass should be replaceable with objects of its subclasses without affecting the correctness of the program.

- Can replace superclass with subclass
- MUST MAINTAIN ALL GUARANTEES!!!



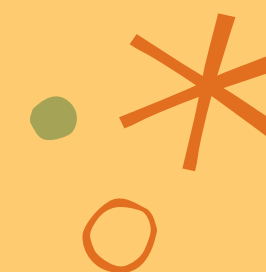


Quick Quiz





Training



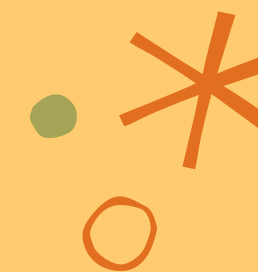


Do Getters Violate Law of Demeter's

And are getters bad
design?

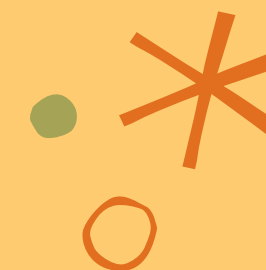


Online Seminar





PATTERNS!!!



Patterns

A common problem?

Evaluate the integrals in Problems 1–100.

1. $\int \frac{1}{\sqrt{x}(1+x)} dx$

3. $\int \sin x \sec x dx$

5. $\int \frac{\tan \theta}{\cos^2 \theta} d\theta$

7. $\int x \tan^2 x dx$

9. $\int x^5 \sqrt{2-x^3} dx$

11. $\int \frac{x^2}{\sqrt{25+x^2}} dx$

13. $\int \frac{1}{x^2-x+1} dx$

15. $\int \frac{5x+31}{3x^2-4x+11} dx$

17. $\int \frac{1}{5+4\cos \theta} d\theta$

19. $\int \frac{\cos x}{\sqrt{4-\sin^2 x}} dx$

21. $\int \frac{\tan x}{\ln(\cos x)} dx$

23. $\int \ln(1+x) dx$

25. $\int \sqrt{x^2+9} dx$

2. $\int \frac{\sec^2 t}{1+\tan t} dt$

4. $\int \frac{\csc x \cot x}{1+\csc^2 x} dx$

6. $\int \csc^4 x dx$

8. $\int x^2 \cos^2 x dx$

10. $\int \frac{1}{\sqrt{x^2+4}} dx$

12. $\int (\cos x) \sqrt{4-\sin^2 x} dx$

14. $\int \sqrt{x^2+x+1} dx$

16. $\int \frac{x^4+1}{x^2+2} dx$

18. $\int \frac{\sqrt{x}}{1+x} dx$

20. $\int \frac{\cos 2x}{\cos x} dx$

22. $\int \frac{x^7}{\sqrt{1-x^4}} dx$

24. $\int x \sec^{-1} x dx$

26. $\int \frac{x^2}{\sqrt{4-x^2}} dx$

27. $\int \sqrt{2x-x^2} dx$

29. $\int \frac{x^4}{x^2-2} dx$

31. $\int \frac{x}{(x^2+2x+2)^2} dx$

33. $\int \frac{1}{1+\cos 2\theta} d\theta$

35. $\int \sec^3 x \tan^3 x dx$

37. $\int x(\ln x)^3 dx$

39. $\int e^x \sqrt{1+e^{2x}} dx$

41. $\int \frac{1}{x^3 \sqrt{x^2-9}} dx$

43. $\int \frac{4x^2+x+1}{4x^3+x} dx$

45. $\int \tan^2 x \sec x dx$

47. $\int \frac{x^4+2x+2}{x^5+x^4} dx$

49. $\int \frac{3x^5-x^4+2x^3-12x^2-2x+1}{(x^3-1)^2} dx$

50. $\int \frac{x}{x^4+4x^2+8} dx$

28. $\int \frac{4x-2}{x^3-x} dx$

30. $\int \frac{\sec x \tan x}{\sec x + \sec^2 x} dx$

32. $\int \frac{x^{1/3}}{x^{1/2}+x^{1/4}} dx$

34. $\int \frac{\sec x}{\tan x} dx$

36. $\int x^2 \tan^{-1} x dx$

38. $\int \frac{1}{x\sqrt{1+x^2}} dx$

40. $\int \frac{x}{\sqrt{4x-x^2}} dx$

42. $\int \frac{x}{(7x+1)^{17}} dx$

44. $\int \frac{4x^3-x+1}{x^3+1} dx$

46. $\int \frac{x^2+2x+2}{(x+1)^3} dx$

48. $\int \frac{8x^2-4x+7}{(x^2+1)(4x+1)} dx$

51. $\int \frac{1}{4+5\cos \theta} d\theta$

Patterns

Has a common solution

$$\int \frac{1}{a^2 - x^2} \frac{1}{2} dx \quad x = a \sin \theta$$

$$\frac{dx}{d\theta} = a \cos \theta$$

$$= \int \frac{1}{(a^2 - a^2 \sin^2 \theta)^{3/2}} (a \cos \theta d\theta)$$

$$= \int \frac{1}{(a^2 - a^2 \sin^2 \theta) \sqrt{a^2 - a^2 \sin^2 \theta}} a \cos \theta d\theta$$

$$= \int \frac{1}{a^2 \cos^2 \theta} d\theta$$

$$= \frac{1}{a^2} \int \sec^2 \theta d\theta$$

$$= \frac{1}{a^2} \tan \theta + C$$

$$\int_0^{\pi/2} \frac{1}{(a^2 - x^2)^{3/2}} dx$$

$$= \frac{1}{a^2} \int_0^{\pi/6} \sec^2 \theta d\theta$$

$$= \frac{1}{a^2} [\tan \theta]_0^{\pi/6}$$

$$= \frac{1}{a^2} \left[\frac{1}{\sqrt{3}} - 0 \right]$$

$$= \frac{1}{\sqrt{3} a^2} \quad \times$$

$$\text{When } x = \frac{a}{2}$$

$$\frac{a}{2} = a \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\text{When } x = 0$$

$$0 = a \sin \theta$$

$$\sin \theta = 0$$

$$\theta = 0$$

$$14) \frac{d}{dx} \ln \cot x$$

$$= \frac{-\operatorname{cosec}^2 x}{\cot x}$$

$$= -\frac{1}{\sin^2 x} \cdot \tan x$$

$$= -\frac{1}{\sin^2 x} \frac{\sin x}{\cos x}$$

$$= -\frac{2}{2 \sin x \cos x}$$

$$= -2 \operatorname{cosec} 2x$$

$$\text{Since } \frac{d}{dx} \ln \cot x = -2 \operatorname{cosec} 2x$$

$$\ln \cot x = \int -2 \operatorname{cosec} 2x dx$$

$$\int \operatorname{cosec} 2x dx = -\frac{1}{2} \ln \cot x$$

$$= \frac{1}{2} \ln \tan x$$

$$\text{Let } x = \frac{\pi}{2}, \frac{dx}{d\theta} = \frac{1}{2}$$

$$\int \operatorname{cosec} \theta \left(\frac{1}{2} d\theta \right) = \frac{1}{2} \ln \tan \frac{\pi}{2}$$

$$\int \operatorname{cosec} \theta d\theta = \frac{1}{2} \ln \tan \frac{\pi}{2} +$$

Patterns

Easy answers!!!

- $\int 1 \, dx = x + C$
- $\int a \, dx = ax + C$
- $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C; n \neq -1$
- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$
- $\int \sec^2 x \, dx = \tan x + C$
- $\int \csc^2 x \, dx = -\cot x + C$
- $\int \sec x(\tan x) \, dx = \sec x + C$
- $\int \csc x(\cot x) \, dx = -\csc x + C$
- $\int \frac{1}{x} \, dx = \ln |x| + C$
- $\int e^x \, dx = e^x + C$
- $\int a^x \, dx = \frac{a^x}{\ln a} + C; a > 0, a \neq 1$
- $\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x + C$
- $\int \frac{1}{1+x^2} \, dx = \tan^{-1} x + C$
- $\int \frac{1}{|x|\sqrt{x^2-1}} \, dx = \sec^{-1} x + C$





Strategy Pattern



Strategy Pattern

The common problem?

Our object needs to use **different variations of an algorithm** or **display different behaviour**, which the object can **switch between during runtime**:

- Changing modes of transportation on Google Maps (walk vs bike vs car)
- Recieving different payment methods (debit card vs PayPal vs cash)



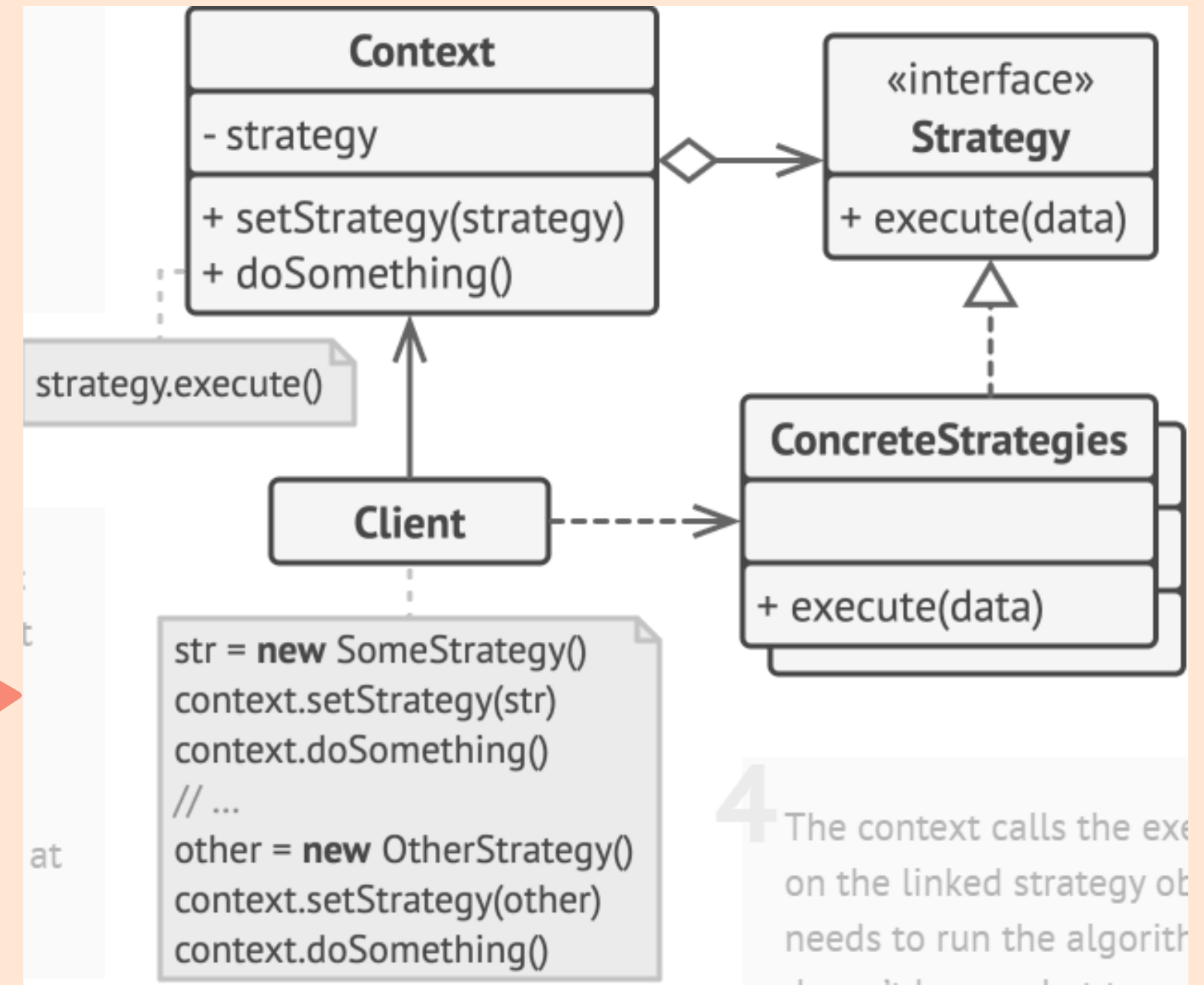
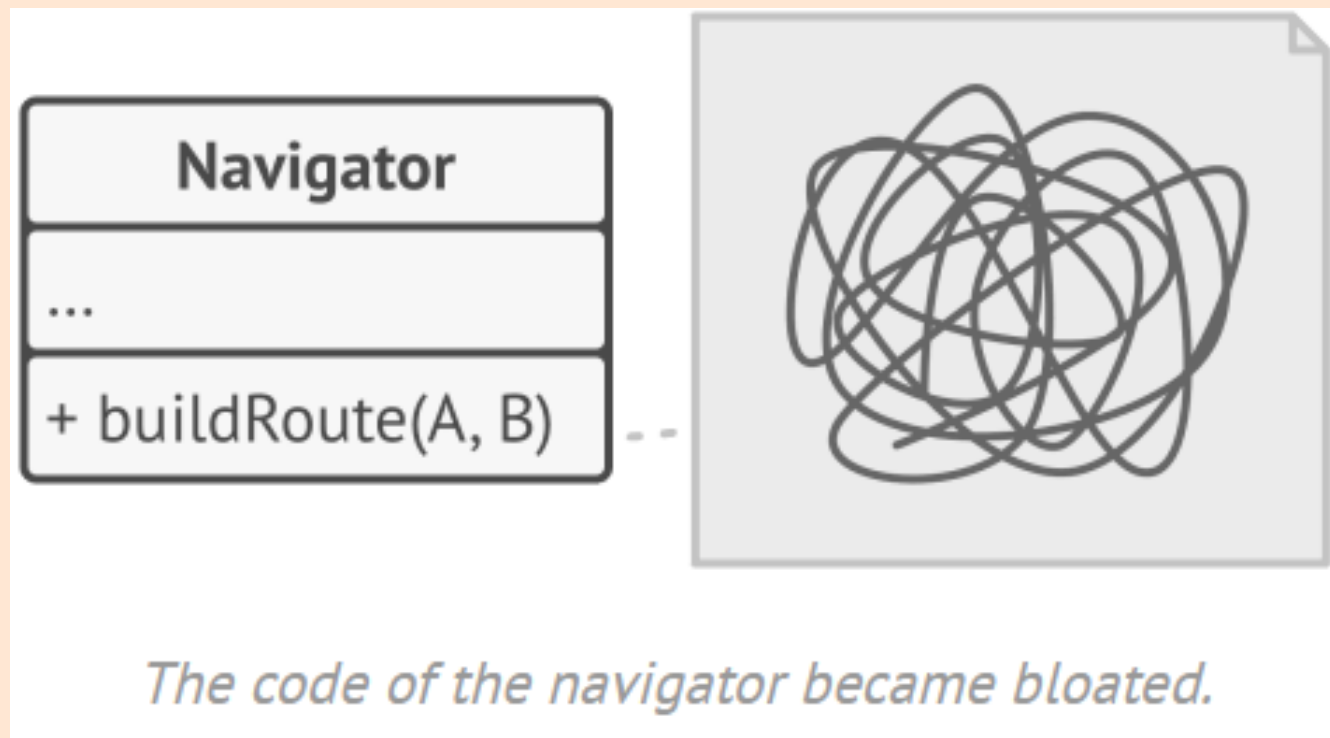
Strategy Pattern

The common solution?

1. Identify the **STRATEGIES**: different behaviours and algorithms the object can switch between.
2. Create Strategy interface with the common method, implemented by each strategy.
3. Add the interface as an attribute to the object using these strategies.

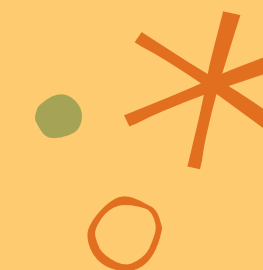


Strategy Pattern





Resturant





Design Principles

How is open/closed principle violated?
How does this make the code brittle?

```
MyFunctionInterfaceA f1 = (x, y) -> x + y  
MyFunctionInterfaceA f2 = (x, y) -> x - y  
MyFunctionInterfaceB f3 = (x, y) -> x > y  
MyFunctionInterfaceC f4 = x -> {  
    double y = 1.5*x;  
    return y + 8.0;  
};  
  
System.out.println( f1.myCompute(10, 20) )  
System.out.println( f2.myCompute(10, 20) )  
System.out.println( f3.myCmp(10, 20) );  
System.out.println( f4.doSomething(10) );
```

Lambdas

Lambdas are “mini functions”, containing:

- parameter list: names only, no types
- arrow token: $\rightarrow$
- body: either single expression or statement block


```
list.stream()  
  .filter( e -> e.length() > 0 )  
  .mapToInt(String::length)  
  .average()  
  .getAsDouble();
```

Streams

A sequence of elements,
pipelined through some
intermediate operations.

- Convert to Stream using
nums.stream()
- Pipe operations with “.”,
nums.stream().op1().op2()
- Convert back to List
using .toList()

The :: Operator

STATIC METHOD

Class::method

- method is static in Class

INSTANCE METHOD

instance::method

- instance of a Class

CLASS CONSTRUCTOR

Class::new

- Calls the Constructor of the Class



App

